Software Engineering Lab 1

NAME: Suhas Mohan Kumar

SRN: PES1UG24AM290

Scenario 1: Self-Service Coffee Kiosk

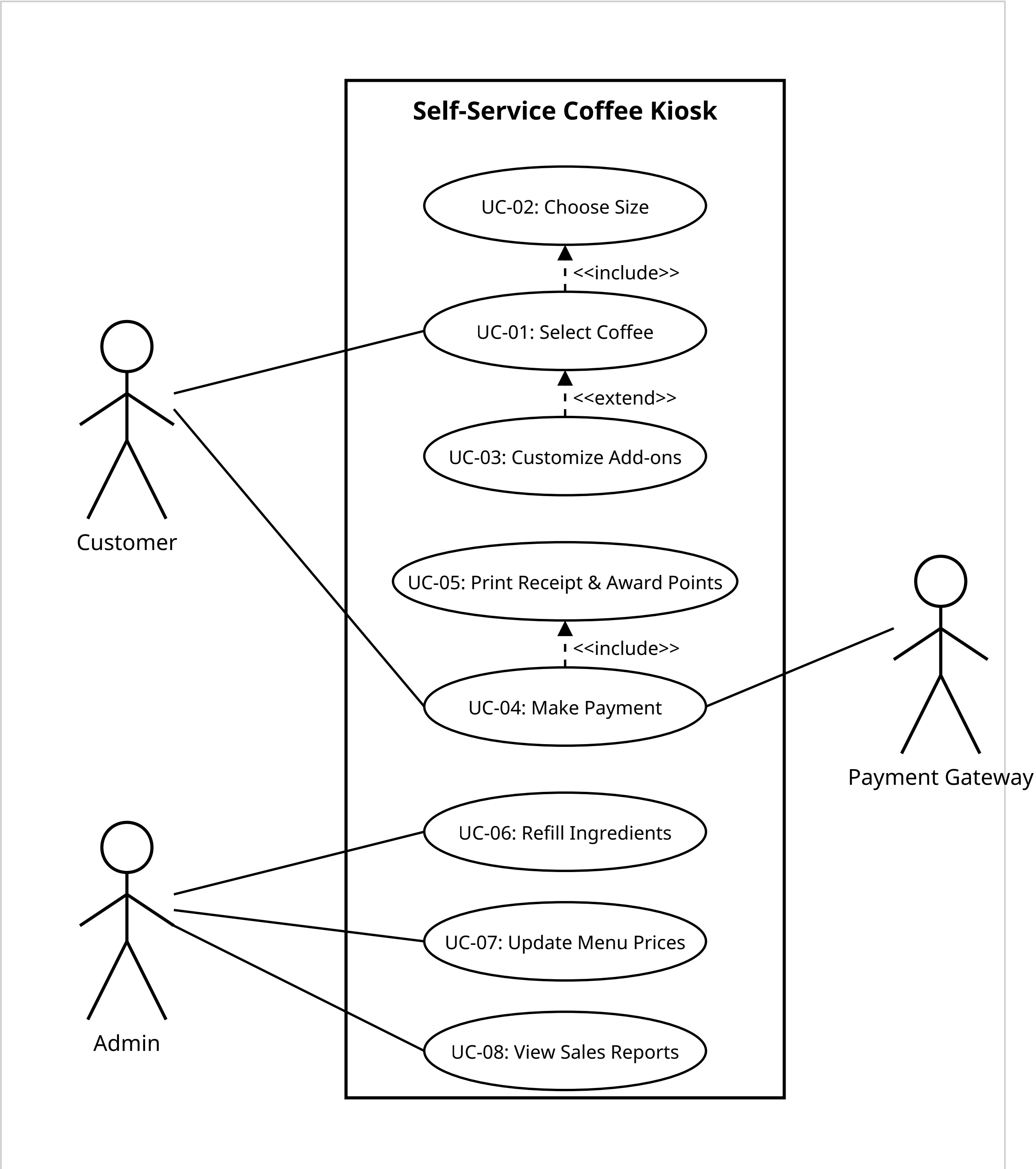
Deliverable 1: Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a customer to select a coffee type.	High	When a customer taps “Espresso,” the screen highlights it and moves to size selection.	Core ordering functionality
FR-002	Functional	The system shall allow the customer to select a drink size.	High	When a size is selected, it is shown as chosen and the next step is prompted.	Required for price calculation and preparation
FR-003	Functional	The system shall allow customers to customize add-ons (extra shot, soy/almond milk, syrups).	Medium	Tapping an add-on highlights the selected item, updates the running total price, and displays the addition under order summary within 1 second.	Enables drink customization to meet customer preferences.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-004	Functional	The system shall process payments via credit/debit card or mobile wallet.	High	Upon payment authorization from the external payment gateway, the system displays a confirmation message on screen within 3 seconds.	Necessary to complete commercial transactions securely.
FR-005	Functional	The system shall print a receipt showing order details and updated loyalty points.	Medium	Following successful payment, a physical or digital receipt is generated listing all ordered items, total cost, and added loyalty points.	Provides proof of purchase and tracks loyalty rewards.
NFR-001	Nonfunctional	The system shall complete any order (selection to payment) in under 60 s.	High	A log shows order completion time is consistently < 60 seconds.	Ensures quick service during peak hours

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Nonfunctional	The system shall provide a password-protected admin mode for café staff.	High	Accessing management features (refill ingredients, update prices, view sales reports) requires entering a valid password; invalid attempts block access with an error message.	Prevents unauthorized access to kiosk settings, inventory, and financial reports.

Deliverable 2: UML Use-Case Diagram



Deliverable 3: Use-Case Flow Document

Use Case Name: Place Coffee Order & Make Payment
Primary Actor: Customer
Secondary Actor: Payment Gateway
Preconditions: The kiosk software is active, on the main home screen, and connected to the network.
Postconditions: Payment is authorized, order details are logged, receipt is printed with updated loyalty points, and the screen resets to default state.

Main Success Scenario

1. Customer selects their desired coffee type (Espresso, Americano, Latte, or Cappuccino) from the touchscreen.
2. System prompts the customer to choose a drink size (Small, Medium, or Large).
3. Customer selects the drink size.
4. System presents customization options for add-ons (extra shot, milk alternative, syrups).
5. Customer adds optional customizations and confirms the order selection.
6. System calculates the total cost and prompts the customer to select a payment method (credit/debit card or mobile wallet).
7. Customer selects their payment method and completes the payment prompt.
8. System connects to the Payment Gateway and sends transaction details for processing.
9. Payment Gateway authorizes the transaction and returns a payment confirmation to the kiosk.
10. System updates the backend database with the new order and customer loyalty points.
11. System prints a physical/digital receipt showing order details and updated loyalty points.
12. System displays an order confirmation message on screen and resets to the main menu state for the next user.

Alternate Flows

- **8a. Payment Declined**
    1. System receives a transaction decline response from the Payment Gateway.
    2. System displays a clear error message on screen: "Payment Declined. Please select an alternative payment method."

3. Customer selects another payment method or inserts a different card.
4. System retries processing through the Payment Gateway and resumes flow at Step 9.
    - *8a1. Exceeded Retries:* If payment fails twice, the system cancels the active transaction, clears the cart, displays a cancellation notice, and returns to the main home screen.

- **5a. Ingredient / Add-on Unavailable**

1. Customer selects an add-on option (e.g., Oat Milk) that is currently out of stock in the backend database.
2. System displays an inline notification: "Selected item is temporarily unavailable."
3. Customer chooses an alternative option or proceeds without the add-on.
4. System updates the order breakdown and resumes flow at Step 6.

Scenario 2: Pharmacy Medicine Pickup System

Deliverable 1: Requirements Table

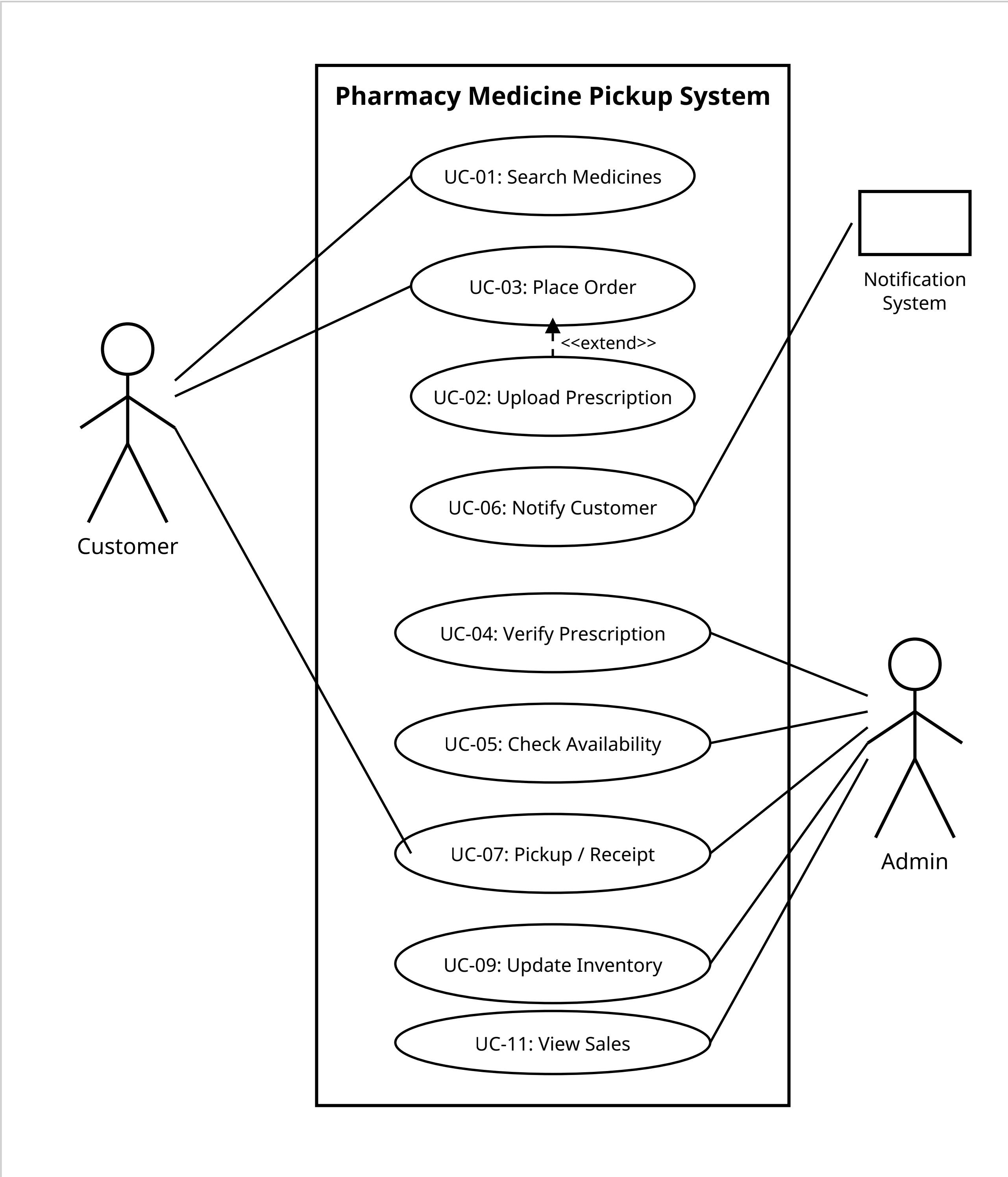
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow customers to search and select medicines.	High	Selected medicines are added to the order summary.	Enables customers to place medicine orders.
FR-002	Functional	The system shall allow customers to upload or scan a prescription for prescription-only medicines.	High	The uploaded prescription is successfully attached to the order.	Required for regulated medicines.
FR-003	Functional	The system shall allow the pharmacist to verify prescriptions and confirm medicine availability.	High	Orders are approved only after prescription verification and stock confirmation.	Ensures legal compliance and accurate dispensing.
FR-004	Functional	The system shall notify customers when their order is ready for pickup.	High	A notification is sent after the pharmacist marks the order as ready.	Keeps customers informed about order status.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall generate a receipt after successful medicine pickup.	Medium	The receipt contains the ordered medicines, pickup date and time, and order ID.	Provides proof of purchase.
NFR-001	Nonfunctional	The system shall allow customers to place an order within 90 seconds.	High	At least 95% of orders are completed within 90 seconds.	Reduces waiting time during busy hours.
NFR-002	Nonfunctional	The system shall provide a responsive and user-friendly interface.	High	Usability testing shows that 95% of users can place an order without assistance.	Improves customer experience.

Deliverable 2: UML Use-Case Diagram & Actors

Use Cases Identified:

- UC-01: Search Medicines
- UC-02: Upload Prescription
- UC-03: Place Order
- UC-04: Verify Prescription (Admin)
- UC-05: Check Medicine Availability (Admin)
- UC-06: Notify Customer
- UC-07: Pickup Medicines
- UC-08: Generate Receipt
- UC-09: Update Medicine Inventory (Admin)
- UC-10: Add/Update Medicines (Admin)
- UC-11: View Sales Reports (Admin)


Pharmacy Medicine Pickup System
UC-01: Search Medicines
UC-03: Place Order
<<extend>>
UC-02: Upload Prescription
UC-06: Notify Customer
UC-04: Verify Prescription
UC-05: Check Availability
UC-07: Pickup / Receipt
UC-09: Update Inventory
UC-11: View Sales
Customer
Notification System
Admin

Deliverable 3: Use-Case Flow Document

Main Success Scenario – Place Medicine Order

1. Customer searches for and selects the required medicines.
2. Customer uploads a prescription, if required.
3. Customer places the order.
4. Pharmacist verifies the prescription.
5. Pharmacist checks medicine availability.
6. Pharmacist prepares the order and marks it as ready for pickup.
7. System sends a notification to the customer.
8. Customer visits the store and collects the medicines.
9. System generates a receipt.
10. Order is marked as completed.

Alternate Flow

- **4a. Invalid Prescription**
  - 4a1. Pharmacist rejects the prescription.
  - 4a2. System notifies the customer that the prescription is invalid or incomplete.
  - 4a3. Customer uploads a valid prescription or contacts the pharmacy.
- **5a. Medicine Unavailable**
  - 5a1. Pharmacist marks one or more medicines as unavailable.
  - 5a2. System notifies the customer about the unavailable medicines.
  - 5a3. Customer may modify or cancel the order.